QUEEN'S UNIVERSITY QUIZ VERSION B
FACULTY OF ARTS AND SCIENCE
SCHOOL OF COMPUTING

CISC 220 001 – Professor Burton Ma
October 05, 9:30AM

INSTRUCTIONS TO STUDENTS:
This quiz is 50 MINUTES in length.
There are 5 questions worth a total of 30 marks in this examination.
Please answer all questions in the question booklet.
No aids are allowed.

Put your name and student number in the space indicated below.

GOOD LUCK!

PLEASE NOTE:
Asking questions during the test is discouraged.
Do your best to answer exam questions as written.

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NAME:

STUDENT NUMBER:

Question 1 (1 mark each, 6 marks total):

Consider a filesystem having the directory structure shown below:

```
???
└── home
    ├── user1
    │   └── courses
    │       ├── CISC204
    │       └── CISC220
    └── user2
        └── courses
            ├── CISC204
            └── CISC352
```

Your answer for each of the following parts should be very brief.

A) The directory indicated by `???` is the root directory of the filesystem. What symbol is used for the name of the root directory?

`/`

B) What is the absolute pathname of the directory `CISC352`?

`/home/user2/courses/CISC352`

C) Suppose that the current working directory is `home`. What single command would you use to list the files in the directories named `CISC220` and `CISC352`?

`ls user1/courses/CISC220 user2/courses/CISC352`

D) Suppose that the current working directory is `CISC204` belonging to `user1`. What single command would you use to copy all the files from the current working directory to the `CISC204` directory belonging to `user2`?

```
cp * /home/user2/courses/CISC204
```

E) Suppose that the current working directory is `courses` belonging to `user1`. What single command would you use to rename the directory `CISC220` to `cisc220`?

```
mv CISC220 cisc220
```

F) Suppose that the current working directory is `courses` belonging to `user2`. What single command would you use to list all the files including the hidden files found in the subdirectories of `courses`?

```
ls -a */
```

Question 2 (1 mark each, 6 marks total):

Your answer for each of the following parts should be very brief.

A) In Linux, a directory is a file that contains what information?

The mapping of filenames to inodes

B) Every command produces an exit status. If a command completes without encountering any errors, what is the expected value of its exit status?

0

C) Suppose that the current working directory has no subdirectories but it does have one or more non-hidden files. What can you conclude about the filenames in the directory if

```
ls *
```

produces the exact same output as

```
ls ?
```

Be as specific as possible in your answer.

All of the filenames are a single character in length.

D) What command could you use to overwrite the contents of a file name `myfile` with the string `OOPS`? Hint: Consider using a redirection.

```
echo OOPS > myfile
```

E) From inside a script, how can you print the number of command line arguments provided to the script?

```
echo $#
```

F) How can you concatenate two strings stored in variables named `str1` and `str2` and store the resulting string in a variable named `str3`?

```
str3=$str1$str2
```

Question 3 (2 marks each, 4 marks total):

A) Recall that if `s` is a variable then `${s:offset:len}` expands to the substring of `s` starting at index `offset` and having length `len`.

Using one or two lines of Bash, how could you print the first half of a string stored in a variable `s`? Assume that the string has even length.

```
n=$(( ${#s} / 2 ))
echo ${s:0:$n}          # $ not required in front of n
```

B) Suppose that you have a file named `classlist` that contains the last name, first name, and student number of each student in a course. A tab character separates the last name from the first name and the first name from the student number. For example:

```
Cade        Barry       3100025
Day         Holly       2999999
Friese      Andy        3001111
Yu          Sue         1990000
```

Write a single pipeline that extracts the student numbers from the file `classlist`, sorts the student numbers in ascending order, saves all the student numbers to a file named `snum` (in sorted order), and prints the first three student numbers (in sorted order). Hint: You need a `tee` somewhere in the pipeline.

```
cut -f3 classlist | sort | tee snum | head -n3
```

Question 4 (4 marks):

Write a script that accepts two command line arguments that are both file names. The first argument is the name of the source file, and the second argument is the name of the destination file.

The script sets the permissions of the destination file to be equal to the permissions of the source file (i.e., it copies the permissions from the source file to the destination file).

No error checking is required in this question. You can assume that two command line arguments are always given, and that the source and destination files exist.

From Assignment 2 recall that the `stat` command can print the permissions of a file:

`stat -c %a` *`filename`*

prints the three-digit octal permissions of the file named *`filename`*.

```
#!/bin/bash

src=$1
dest=$2

# permissions of src
perm=$(stat -c %a “$src”)       # quotes required

chmod $perm “$dest”             # quotes required
```

Question 5 (10 marks):

The fact that file removal in Linux is permanent can be inconvenient. Instead of removing a file, we can create a command that moves the file into a hidden trashcan directory.

Create a script named `trash.sh` that moves a single file into a hidden trashcan directory by performing the following steps:

1. finds the name of the directory `dir` that the file is located in
2. creates a directory named `.trash` in `dir`
3. changes the permissions of the `.trash` directory so that the owner has all permissions and all other users have no permissions
4. prints the message "`moving` *`filename`* `to the trashcan`" where *`filename`* is the first command line argument provided to the script (see below)
5. moves the file into the `.trash` directory created in Step 2

Obviously, the script requires one command line argument (the pathname of the file to move into the trash).

No error checking is required. Assume that one command line argument is always provided to the script, and that the argument is the pathname to a file that is owned by the user of the script. Also, assume that the pathname always includes the directory that the file is located in.

Step 2 will cause an error message to be printed if the directory `.trash` already exists in `dir`. You should ignore the fact that this error can occur.

Hint: If `str` is a variable then `${str%pattern}` expands to the string formed by removing the shortest trailing part of `str` that matches `pattern`.

Example 1:

```
trash.sh ./myfile
```

moves `./myfile` into the `.trash` directory created in the current working directory

Example 2:

```
trash.sh /home/user1/somedir/myfile
```

moves `/home/user1/somedir/myfile` into the `.trash` directory created in the directory `/home/user1/somedir/`

```
#!/bin/bash

pathname=$1

# Step 1
dir=${pathname%/*}

# Step 2
trashdir=$dir/.trash
mkdir “$trashdir”

# Step 3
chmod u=rwx,go= “$trashdir”       # chmod 700 “$trashdir”

# Step 4
echo “moving $pathname to the trashcan”

# Step 5
mv “$pathname $trashdir”
```